

Group Theory

1.

- (a) There are two different isomorphism classes of groups of order 12 which can be written as $\mathbb{Z}/6\mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z}$. What are they?
- (b) Suppose G is a group with a normal subgroup N , and let $\pi: G \rightarrow G/N$ be the projection to the quotient. Prove that G is a(n internal) semi-direct product $N \rtimes G/N$ if and only if there exists a homomorphism $s: G/N \rightarrow G$ such that $\pi \circ s = \text{id}: G/N \rightarrow G/N$. That is to say, if and only if the exact sequence of groups

$$1 \rightarrow N \rightarrow G \rightarrow G/N \rightarrow 1$$

splits on the right.

1. Solution

- (a) Up to isomorphism they are $\mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ and D_6 .
- (b) Suppose that $G = N \rtimes G/N$, then the map $G/N \rightarrow N \rtimes G/N$ which sends $x \mapsto (e, x)$ is the desired splitting map. On the other hand, suppose such a splitting exists, then we need to show that $Ns(G/N) = G$ and that $N \cap s(G/N) = \{e\}$. For the first part, the section map s sends a coset to a particular representative for that coset Nh . Then, since every $g \in G$ is in some coset, we get that every g can be written as nh for some $n \in N$ and $h \in s(G/N)$. For the second part, if $g \in N \cap s(G/N)$, then write $g = s(h)$. Then $\pi(g) = e$ since $g \in N$, but $\pi(g) = \pi \circ s(h) = h$, so $h = e$ and thus $g = s(e) = e$.

2. (August 2018 Problem 2) For a finite group G , denote by $s(G)$ the number of subgroups of G .

- (a) Show that $s(G)$ is finite.
- (b) Show that if H is a nontrivial normal subgroup of G , then $s(G/H) < s(G)$.
- (c) Show that $s(G) = 2$ if and only if G is cyclic of prime order.
- (d) Show that $s(G) = 3$ if and only if G is cyclic of order p^2 for a prime p .

2. Solution

- (a) Certainly every subgroup is also a subset, and there are at most $2^{|G|}$ subsets of G , so $s(G) \leq 2^{|G|} < \infty$.
- (b) By the lattice isomorphism theorem, the subgroups of G/H are in bijection with subgroups of G containing H . Certainly the number of subgroups of G containing H is at most the number of subgroups of G , and since H is nontrivial we know the trivial subgroup does not contain H , which makes the inequality strict.
- (c) Certainly, if G is cyclic of order p , then its only subgroups are $G, \{e\}$. On the other hand, suppose $s(G) = 2$. Since $s(G) > 1$, G is not trivial. Let p be a prime dividing $|G|$,

then by Cauchy's theorem there is a $g \in G$ of exact order p . But then $\langle g \rangle$ and $\{e\}$ are 2 subgroups of G , so we have already found all the subgroups of G .

- (d) Certainly, if G is cyclic of order p^2 generated by g , then its only subgroups are $G, \langle g \rangle, \{e\}$. On the other hand, suppose $s(G) = 3$, so that G has exactly 1 nontrivial proper subgroup. As before, we can find an element g of exact order p . Furthermore, p is the only prime which divides $|G|$, as if another prime q divided $|G|$ we would have another nontrivial proper subgroup, but we know G has a unique such subgroup. Thus, $|G| = p^n$ for some n . Now, let $h \in G \setminus \langle g \rangle$, then h generates a nontrivial subgroup, so that subgroup cannot be proper, so $\langle h \rangle = G$, and G is cyclic. Since G is cyclic of order p^n , it has $n + 1$ subgroups, generated by h^{p^k} for $k \leq n$. Thus, $n + 1 = 3$, so $n = 2$.

3. (January 2018 Problem 4) Let G be a finite group. Denote by $\text{Aut}(G)$ the group of automorphisms of G , and by $Z(G) \subset G$ the center of G .

- (a) (Slightly modified) Let $\text{Inn}(G) \subset \text{Aut}(G)$ be the subgroup consisting of automorphisms coming from conjugation, i.e. automorphisms of the form $x \mapsto gxg^{-1}$ for some $g \in G$. Prove that $\text{Inn}(G)$ is isomorphic to $G/Z(G)$.
- (b) Show that if $G/Z(G)$ is cyclic, then G is abelian.
- (c) Show that if $\text{Aut}(G)$ is cyclic, then G is abelian.
- (d) Show that if G is abelian, then $x \mapsto x^{-1}$ is an automorphism of G . (In fact this is an if and only if.)
- (e) Deduce that there is no group G such that $\text{Aut}(G)$ is a nontrivial cyclic group of odd order.

3. Solution

- (a) Denote by $\text{conj}_g(x)$ the “conjugation by g ” automorphism, $\text{conj}_g: G \rightarrow G$, $\text{conj}_g(x) = gxg^{-1}$. Define a map $G \rightarrow \text{Inn}(G)$ by $g \mapsto \text{conj}_g$. We will show that the kernel of this map is equal to $Z(G)$. Certainly, if $z \in Z(G)$, then $zxz^{-1} = x$ for all x , so $\text{conj}_z = \text{id}$. On the other hand, suppose $\text{conj}_g = \text{id}$, that is, that $gxg^{-1} = x$ for all x . Then multiplying on the right by g gives us $gx = xg$ for all x , so $g \in Z(G)$.
- (b) Suppose $G/Z(G)$ is cyclic with generator $gZ(G)$. Then any $x, y \in G$ can be written uniquely as $x = g^{n_1}z_1$, $y = g^{n_2}z_2$, for $z_1, z_2 \in Z(G)$. But then

$$xy = g^{n_1}z_1g^{n_2}z_2 = g^{n_1}g^{n_2}z_1z_2 = g^{n_2}g^{n_1}z_2z_1 = g^{n_2}z_2g^{n_1}z_1 = yx,$$

and G is abelian.

- (c) If $\text{Aut}(G)$ is cyclic, then its subgroup $\text{Inn}(G) \cong G/Z(G)$ is also cyclic.
- (d) Notice that $(xy)^{-1} = y^{-1}x^{-1} = x^{-1}y^{-1}$ since G is abelian.
- (e) By the proceeding, if $\text{Aut}(G)$ is nontrivial and cyclic, then G is abelian, but then $x \mapsto x^{-1}$ is an element of $\text{Aut}(G)$ of order 2, so $2 \mid |\text{Aut}(G)|$. Thus, it is impossible for $\text{Aut}(G)$ to be nontrivial cyclic of odd order.

4. (January 2017 Problem 1) Give an example of each of the following.

- (a) A group G with a normal subgroup N such that G is not a semidirect product $N \rtimes G/N$.
- (b) A finite group G that is nilpotent but not abelian.
- (c) A group G whose commutator subgroup $[G, G]$ is equal to G .
- (d) A non-cyclic group G such that every Sylow subgroup of G is cyclic.
- (e) A transitive action of S_3 on a set X of cardinality greater than 3.

4. Solution

- (a) The smallest such example is $G = \mathbb{Z}/4\mathbb{Z}$, $N = 2\mathbb{Z}/4\mathbb{Z}$. Then $G/N \cong \mathbb{Z}/2\mathbb{Z}$, but there is only one nontrivial map $\mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$, and it is not a splitting for the quotient map. Therefore, $\mathbb{Z}/4\mathbb{Z}$ is not a semidirect product of $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z}$. A non-abelian example would be the quaternion group Q with subgroup $N = \{\pm 1\}$. An infinite example would be $\mathbb{Z}$ with subgroup $N = 2\mathbb{Z}$.
- (b) The smallest examples are D_4 and Q , the quaternion group.
- (c) The smallest such example is $G = A_5$, which is simple and therefore its commutator subgroup must be equal to the whole group.
- (d) Again, take $G = S_3$, or D_n for n odd.
- (e) Consider the action of S_3 on itself by left multiplication.

5. Prove that any nilpotent group is solvable.

5. Solution Suppose G is nilpotent of nilpotency class n . Write the upper central series of G :

$$Z_0 = \{e\} \triangleleft Z_1 = Z(G) \triangleleft Z_2 \triangleleft \cdots \triangleleft Z_n = G.$$

Then we have that in this normal series $Z_{i+1}/Z_i = Z(G/Z_i)$, so in particular the factor group Z_{i+1}/Z_i is abelian. Hence, G is solvable. (You can also do this problem with the upper central series, or indeed with any central series.)

6. Give an example of a group G with subgroups $N \supseteq H$ so that:

- N is normal in G ,
- H is normal in N ,
- H is *not* normal in G .

6. Solution Let $G = D_4 = \langle r, t \rangle$ the dihedral group of order 8, with $r^4 = t^2 = e$ and $trt = r^{-1}$. Let $N = \langle r^2, t \rangle$ and let $H = \langle t \rangle$. Then N is normal in G since N has index 2, H is normal in N since N is abelian, but H is not normal in G because $rtr^{-1} = r^2t \neq t$.

7. (August 2021 Problem 3) Let G be a finite group. For given $n > 1$, we say that G admits roots of degree n if, for every $x \in G$, there exists $y \in G$ such that $y^n = x$.

- (a) Show that if n is coprime to $|G|$, G admits roots of degree n .
- (b) Conversely, prove that if G admits roots of degree n , then n is coprime to $|G|$.

7. Solution

- (a) We will prove the proposition in the case where G is cyclic, and the result will follow because each $x \in G$ lives in the cyclic subgroup $\langle x \rangle$. So, let G be cyclic and generated by x , and consider the group homomorphism $\phi_n : G \rightarrow G$ given by $x \mapsto x^n$. We want to show that this map is surjective. Since G is finite, it is enough to show that this map is injective, and hence it is enough to show that $\ker \phi_n = \{e\}$. Suppose $x \in \ker \phi_n$, then $x^n = e$, so the order x , $\text{ord } x$, divides n . But also $\text{ord } x$ divides $|G|$. Since n is coprime to $|G|$ we must have that $\text{ord } x = 1$, i.e. $x = e$. Thus, $\ker \phi_n$ is trivial.
- (b) We will prove this by contradiction: suppose $\gcd(n, |G|) > 1$ and G admits roots of degree n . If G admits roots of degree n , it also admits roots of degree p for every $p \mid n$, so it suffices to consider the case where $n = p$ for some prime p dividing $|G|$. Let $p \mid |G|$, and let $g \in G$ generate a cyclic subgroup of order p^k which is not contained in any cyclic subgroup of order p^{k+1} . By Cauchy's theorem, there a cyclic subgroup of order p , and we are just taking a maximal cyclic p -group containing such a subgroup. Since G admits roots of degree p , we can find $h \in G$ with $h^p = g$. First, notice that for any k , $(g^k)^p$ lies in a subgroup of index $\geq p$ inside $\langle g \rangle$, and therefore $h \notin \langle g \rangle$. But, on the other hand, $\text{ord } h = p^{k+1}$, and by construction g is not contained in any cyclic group of order p^{k+1} , a contradiction. Therefore, G cannot admit roots of degree p .

8. (August 2019 Problem 5) Let G be the group of 2×2 invertible upper triangular matrices over the field $\mathbb{F}_p$.

- (a) Prove that G has only one subgroup of order p , and that this subgroup is normal. (Hint: first show that an element of order p must have 1's on the diagonal.)
- (b) Prove that G is solvable by exhibiting a homomorphism f from G to an abelian group A such that the kernel of f is also abelian.
- (c) Prove that if $p \neq 2$, G is not nilpotent.

8. Solution

- (a) Given a matrix

$$A = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix},$$

we have that the diagonal entries of A^n are a^n and c^n . Thus, if A has order p , then $a^p = c^p = 1$, but by Fermat's Little Theorem, $a^p = a$, and so $a = c = 1$. Thus, the p

matrices of the form

$$\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$$

form the only subgroup of order p . You can show it is normal either by conjugating by an arbitrary element of G , or else by computing $|G|$ and arguing that this is the unique Sylow- p subgroup of G , hence is normal.

- (b) Let the subgroup from part (a) be U , and let T be the subgroup of diagonal matrices. Then $U \cap T = \{I\}$, and we can write any matrix

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & c \end{pmatrix} \begin{pmatrix} 1 & a^{-1}b \\ 0 & 1 \end{pmatrix}$$

where the first factor is in T and the second is in U . Thus, $G = U \rtimes T$, and so G/U is isomorphic to T , which is abelian. Thus, $f: G \rightarrow G/U$ is the homomorphism we want.

- (c) Assume $p \neq 2$. This is probably easiest to show using the lower central series. Notice that the diagonal entries of any commutator $ABA^{-1}B^{-1}$ are going to be 1, so actually every commutator falls inside U . So, $[G, G]$ is a subgroup of U , and it isn't the trivial subgroup since G is not abelian, so the only other option is $[G, G] = U$. But now, the second term of the lower central series $G_2 = [G, U]$ is still equal to U , since for example the matrices

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

do not commute (recall that we've assumed $p \neq 2$). Thus, $ABA^{-1}B^{-1}$ is a nontrivial commutator in $[G, U]$, so $[G, U]$ is a nontrivial subgroup of U , hence is all of U .

9. (January 2020 Problem 5)

- (a) Find a nontrivial finite subgroup of $\mathrm{GL}_2(\mathbb{R})$.
- (b) Find an infinite, abelian, but non-cyclic subgroup of $\mathrm{GL}_2(\mathbb{R})$.
- (c) Find a proper, normal subgroup of $\mathrm{GL}_2(\mathbb{Z})$ of finite index.

9. Solution

- (a) Consider the subgroup of order 2 $\{I, -I\}$.
- (b) Consider the subgroup consisting of diagonal matrices.
- (c) For each prime p there is a map $\mathrm{GL}_2(\mathbb{Z}) \rightarrow \mathrm{GL}_2(\mathbb{F}_p)$ given by reducing each entry mod p . The kernel of such a map must be proper and normal, and since the target is a finite group it must be of finite index.